Singapore School Kinderland Secondary-2

Computer Science — Term 2 Comprehensive Practice Worksheet

Chapter 1.3 (Storage & Compression) & Chapter 2.2, 2.3 (Transmission & Security)

[TEACHER ANSWER KEY]

Date: _____ **Student Name:** TEACHER ANSWER KEY & MARK SCHEME

Class: Secondary 2 (Sec 2) **Total Marks: 40 Marks | Obtained: 40 / 40**

SECTION 1: DATA STORAGE & FILE SIZE CALCULATIONS [10 MARKS]

1. Unit Conversions: (3 Marks)

Convert the following storage capacities into the requested units:

a) Convert 4 KiB into Bytes: **4 × 1024 = 4,096 Bytes [1 mark]**

b) Convert 2.5 MiB into KiB: **2.5 × 1024 = 2,560 KiB [1 mark]**

c) Explain the difference between 1 Kilobyte (KB) and 1 Kibibyte (KiB):

Model Answer: 1 KB uses SI decimal system (10^3 = 1,000 Bytes), whereas 1 KiB uses IEC binary system (2^10 = 1,024 Bytes). [1 mark]

2. Image File Size Calculation: (3 Marks)

A bitmap image has a resolution of **1024 × 768 pixels** and a colour depth of **16 bits per pixel**.

Calculate the file size in **Kibibytes (KiB)**. Show your full working.

Working:

1. Size in bits = 1024 × 768 × 16 = 12,582,912 bits. [1 mark]

2. Size in Bytes = 12,582,912 ÷ 8 = 1,572,864 Bytes. [1 mark]

3. Size in KiB = 1,572,864 ÷ 1024 = 1,536 KiB (or exactly 1.5 MiB). [1 mark]

3. Audio File Size Calculation: (4 Marks)

A studio records a **40-second stereo audio clip** with a sampling rate of **44,100 Hz** and a sample resolution of **16 bits**.

Calculate the size of the uncompressed audio file in **Bytes** and in **Mebibytes (MiB)**.

Working:

1. Bits = Sample Rate × Resolution × Duration × Channels

= 44,100 × 16 × 40 × 2 (stereo) = 56,448,000 bits. [1 mark]

2. Bytes = 56,448,000 ÷ 8 = 7,056,000 Bytes. [1 mark]

3. Size in MiB = 7,056,000 ÷ (1024 × 1024) = 7,056,000 ÷ 1,048,576 ≈ 6.73 MiB. [2 marks]